ELECTROCHEMICAL TECHNIQUES

INTRODUCTION

Electrochemistry, in contrast to most materials characterization disciplines, contains a bifurcated methodology having applications both in materials synthesis and as a physical characterization technique. On the one hand, a variety of functional surfaces are prepared electrochemically, for example, the anodization of aluminum, electroetching of semiconductors, and electroplating of metallic films. Similarly, the electrosynthesis of bulk materials such as conducting organic polymers has taken on importance in recent years. On the other hand, electrochemistry has been demonstrated to be of use as a physical characterization technique for the quantification of conducting and semiconducting surfaces. The corrosion properties of metallic surfaces and semiconductor electronic properties are two key areas where electrochemical characterization has had a high impact. In keeping with the central analytical theme of this work, this chapter limits its focus to the class of electrochemical techniques specifically aimed at characterizing materials and their interfaces (as opposed to synthetic electrochemistry). In some cases, the line between materials synthesis and measurement can become hazy. The field of sensors based on chemically modified electrodes (CMEs) is a good example of this confusion. Often sensors of this type are electrosynthesized and the synthesis parameters represent an important portion of the systems characterization. Cases such as this will be included in the topical coverage of this chapter.

Electrochemistry differs from other analytical techniques in a second important way. Unlike most instrumental characterization techniques, the sample under study is made into part of the measuring circuitry, and thus, inappropriate sample preparation can effectively lead to a malfunction of the instrument. More importantly, this synergism demands that the user have some knowledge of the detailed circuitry associated with the instrument being employed. Most instruments can be treated as "black boxes"; the user need only understand the rudiments of the instrument's functions. However, if this approach is employed in the execution of an electrochemical experiment, the result is often an artifactual instrument response that is easily misinterpreted as the chemical response of the sample. The electrochemical literature repeatedly testifies to the preeminence of this problem.

Modern electroanalytical techniques have become viable characterization tools because of our abilities both to precisely measure small currents and to mathematically model complex heterogeneous charge transfer processes. Modeling of electrode-based charge transfer processes is only possible using digital simulation methods. The mathematics and computer science related to this underpinning of electrochemistry is sophisticated and beyond the scope of this work. In most cases, an incomplete understanding of this aspect of electroanalysis will not affect the quality of the experimental outcome. In cases where the experimenter wishes to obtain a more detailed understanding of the physical chemical basis of electrochemistry a variety of texts are available. Two volumes, the first by Bard and Faulkner (1980) and the second by Gileadi (1993) are particularly well suited to a more detailed understanding of electrochemistry and electroanalysis. The latter volume is specifically aimed at the materials scientist while the former work is one of the central texts in physical electrochemistry. A third text edited by Kissinger and Heineman (1996) is an excellent source of experimental details and analytical technologies. All of these texts will be most valuable as supplements to this work, providing more chemical details but less materials-characterization emphasis than found here.

Electrochemical measurements can be divided into two categories based on the required instrumentation. Potentiometric measurements utilize a sensitive voltmeter or electrometer and measure the potential of a sample against a standard of known potential. Voltammetric measurements utilize a potentiostat to apply a specified potential waveform to a sample and monitor the induced current response. Amperometric measurements, in which the potential is held constant and the current monitored as a function of time, are included in this latter category. Potentiometric measurements form the historical basis of electrochemistry and are of present utility as a monitoring technique for solution-based processes (i.e., pH monitoring or selective ion monitoring). However, potentiometry is limited with regard to materials characterization. Thus, this chapter focuses on voltammetric measurements and the use of the potentiostat. The potentiostat is fundamentally a feedback circuit that monitors the potential of the test electrode (referred to as the working electrode) versus a reference half-cell (typically called a reference electrode). If the potential of the working electrode drifts from a prescribed offset potential versus the reference electrode, a correcting potential is applied. A third electrode, the counterelectrode (or auxiliary electrode) is present in the associated electrochemical cell to complete the current pathway. The potentiostat typically contains a current-following circuit associated with the auxiliary electrode, which allows a precise determination of the current, which is reported to a recording device as a proportional potential. The rudimentary operation of the potentiostat is covered in this text. However, for many applications a more detailed analysis of potentiostat electronics is desirable. To this end, the reader is directed to Gileadi's excellent monograph (Gileadi et al., 1975).

Potentiostats are available from a series of vendors and range in price from less than $1000 to approximately